SPPA-MVFit Supplement: Protocol, Reproduction, and Secondary Evidence

Purpose

This supplement records the confirmatory protocol path and reproduction commands, the external neural-wave protocol, the open-label probe, the real-image probes and replay audits, the role-preservation diagnostic, the descriptor and Unreal runtime tests, and the full tables behind the figures of the main paper. It also retains material moved from the main text during the mission reframe: the end-to-end compile algorithm (Section S.1), the real-image probe grid (Section S.4), the packaged-runtime scaling figure (Section S.8), and the secondary, surface-metric, OBB, fitting-sequence, external-gallery, and real-stream localization material (Section S.9).

Primary endpoint (held-out)

- Estimand: clean 64-cubed voxel IoU of `sppa_mvfit` minus `generic_mvfit`. Units: 240 source actors (12 family$\times$stratum cells, equal strata).
- Decision: stratified bootstrap 95% CI lower bound $> +0.030$. Result: mean $\Delta = 0.190$, CI $[0.181, 0.199]$, **H1 PASS**.
- Strata: CSG-ID 0.209, implicit-OOD 0.172 (both positive). Provenance: `synthetic_geometry` only.

Reproduction commands

Run from the git repository root (or the `paper_semantic_proxy_3d` junction directory inside it). Do not re-optimize the held-out methods after inspection.

Table S1: Minimal reproduction checks for the primary claim.

Command	Purpose	Expected
`python-mpytestpaper_semantic_proxy_3d/reproducibility/sppa_mvfit/tests/test_contract.py`	Shared builder, budget, no cross-imports, split balance.	7 passed
`pythonpaper_semantic_proxy_3d/reproducibility/sppa_mvfit/benchmark/verify_package.py--development`	Development package integrity.	PASS
`pythonpaper_semantic_proxy_3d/tools/reproduce_sppa_mvfit_paper.py--strict`	Hash/gate checks for the submission snapshot.	0 blockers

Held-out generation/evaluation commands exist in `reproducibility/sppa_mvfit/README.md` and require the protocol PASS record, pretest freeze, and NIST-bound seed manifest. Re-running method optimization for a better score is forbidden by protocol. Main-paper tables are regenerated from the released `raw_metrics.csv` by `export_paper_tables.py`.

Protocol bookkeeping and artifact integrity

The margin (+0.030), the power analysis, and the observation conditions were fixed in writing before evaluation (protocol Amendment 01). Seeds were derived from a NIST Randomness Beacon pulse after a pre-test freeze and a written triple-role protocol release (Amendment 03); the external neural wave was registered as Amendment 05 before any measurement. SHA-256 manifests for the pretest freeze, the protocol PASS record, the prediction binary, and the raw metrics are stored with the artifact package under `reproducibility/sppa_mvfit/`.

Claim boundaries

Supported: family-conditioned occupancy gain under synthetic calibrated silhouettes; shared generator; optional runtime contract evidence. Not supported: measured flight, real-UAV 3D ground-truth accuracy, operator benefit, universal open-set geometry, neural image-to-3D SOTA ranking, or transfer of the preregistered occupancy margin to external real meshes (not replicated at n=52 in the post-hoc sanity check, main-text Section 4.7). Real-image figures in this supplement remain qualitative probes without 3D ground truth and use declared replay assumptions where telemetry is shown.

S.1 Representation Design Space

The relevant comparison is not a single "3D representation" baseline. A UAV digital twin can visualize a semantic detection through several families of representations, each making different assumptions about input evidence, runtime budget, geometry, and operator interpretation. Geometry-only displays such as boxes, ellipsoids, or billboards remain useful lower-bound controls, but they are not the scientific competitor. SPPA targets the gap between a generic box and a curated asset: a role-labeled primitive actor when only sparse semantic telemetry is available.

Table S2 states the resulting deployment policy. It deliberately gives priority to curated assets and mapping geometry when those are the correct tools. The semantic-label-to-parts mechanism behind that policy is Figure 3 of the main text (restored there in the 2026-07-21 readability pass; an identical copy of that graphic was kept here until then), and Figure S1 writes the end-to-end compile chain out in exact steps.

Algorithm 1 — SPPA compile: from one detection to a displayed actor.

Input: observation x_t (track i, label ℓ, confidence c, bounding box b, mask m, camera/flight geometry, pose estimate ξ_t).
Output: one `SPPA-DESC-0.2` descriptor per track plus per-frame `SPPA-UPD-0.2` updates.

1. *Normalize*: map ℓ through the reviewed ontology to `exact_class`, `keyword_archetype`, or `fallback_unknown_label`.
2. If fallback: emit the generic uncertainty-aware volume and skip to step 8—an unknown label is never hallucinated into a specific object.
3. Select the frozen family graph G_a: eight slots, each a primitive with role, transform, and dimension priors (Figure 2).
4. *Evidence* (detector/image path only): (4.1) **Observe**—store the mask and camera geometry in a `SPPA-OBS-0.1` record; (4.2) **Project**—project the mask to an oriented ground footprint and keep its PCA axial orientation (defined modulo π); (4.3) **Scale**—turn the footprint into a metric length/width/height proposal, only when metric calibration is declared; (4.4) **Gate**—clamp implausible proportions through the family aspect prior and withhold unsafe image-derived pose or yaw.
5. Parameters: $\theta = \theta_0$ (tag/text path) or fitted θ^* (MVFit, Figure 5).
6. Build(g, θ): instantiate the eight role-labeled primitives plus connector cylinders between semantically attached parts.
7. Emit the compact update descriptor once per track ($\approx$1.45 kB); the renderer builds the actor from it.
8. Per frame: apply pose/material/uncertainty updates in place; topology is regenerated only when the class or archetype changes.

Figure S1: The end-to-end SPPA pipeline in exact steps (moved from the main text). Both input contracts (tag/text and detector/image) terminate in the same deterministic builder, so their outputs are directly comparable. No language model or neural generator runs at runtime; the offline-drafted recipes are reviewed and versioned.

Table S2: Operational representation policy for semantic detections.

Situation	Preferred representation	Reason	Boundary
Matched project asset exists	Curated asset backend	best appearance and lower measured cost in the cow/tower subset	asset library must match class and scale
Known semantic family, no asset	SPPA primitive actor	persistent, local, role-labeled 3D proxy from label/pose input	finite reviewed archetypes
Unknown or adversarial label	SPPA fallback volume	avoids inventing a specific object from weak semantics	not visually specific
Clean crop plus offline budget	Image-to-3D or asset authoring	richer mesh can be generated outside the flight update loop	no per-frame track contract
Planning or occupancy task	Occupancy / ESDF / map layer	represents navigable space and collision evidence	not a role-labeled display actor

S.2 External Neural Wave: Protocol Details and Exclusions

Protocol. Sixty held-out test actors were selected deterministically (five per morphology-by-stratum cell, lexicographic `case_id` order). The measured generators were TripoSR [2] (`stabilityai/TripoSR`) and Hunyuan3D-2mini-turbo [3] (`tencent/Hunyuan3D-2mini`, checkpoint `hunyuan3d-dit-v2-mini-turbo`, octree 380, 5 sampling steps, FlashVDM), joined in a post-hoc flagship extension (run `20260719_flagship_wave`) by TripoSG [1] (local

VAST-AI/TripoSG 1.5B rectified-flow weights, 50 steps, guidance 7.0) and the flagship Hunyuan3D-2 full model [3] (`hunyuan3d-dit-v2-0`, 50 steps, guidance 5.0, octree 380, FlashVDM); all four run at fixed seed 12345. Each neural generator received two prespecified inputs: (a) a clean-crop shaded oblique render of the source actor, the generator's best input condition, and (b) the actual 96×96 top telemetry mask as the input image, the closest visual analog of the channel SPPA consumes. Outputs were aligned to ground truth by a frozen frame convention (a one-time 48-candidate signed-permutation search fitted on twelve calibration cases; any selection bias this introduces can only inflate neural IoU, so the direction is conservative for SPPA), uniform scale to the ground-truth bounding box, and ground-truth bounding-box center and yaw frame; no per-case tuning was allowed. Voxel IoU was measured at 64^3 with the same voxelizer and WORLD frame. Two hard crashes (2/480 runs, inherited from the Hunyuan3D-2mini-turbo oblique condition; the two flagship generators completed 120/120 runs without a crash) are reported as exclusions; SF3D (install/build failure under the Amendment 05 Python 3.12 stack, after an earlier Python 3.10 attempt produced no benchmark event before a 20-minute timeout), SPAR3D (gated weights), and TRELLIS.2 (no Windows environment) remain documented environment-level exclusions, so no measured quality result exists for them in either attempt. SPPA-MVFit, Generic-MVFit, and visual-hull rows are the existing confirmatory clean-condition results on the same sixty cases.

Input parity and the TripoSG mask artifact. Every generator received the input PNG as-is, with no background removal or bounding-box crop, matching the TripoSR/Hunyuan convention of the first wave. TripoSG's official pipeline expects a background-removed RGBA input, so its mask-condition score (0.002) is an input-parity artifact of that choice, not a measured property of the model; it is kept in the table for completeness and excluded from every range quoted in the main text. The parity rule and its consequence are registered in the run manifest (`benchmarks/neural_external_wave/runs/20260719_flagship_wave/MANIFEST.md`).

Table S3: Measured external neural wave (Amendment 05, secondary; extended post-hoc with the two flagship generators). Mean voxel IoU at 64^3 on the same sixty held-out cases (Hunyuan3D-2mini-turbo (a): n=58 after the two reported hard crashes), with parenthesized paired ΔIoU versus SPPA-MVFit. Triangles and payload are mesh-output means; generation time is warm per-case wall time on the same RTX 5090; VRAM is uncapped peak CUDA allocation. SPPA rows are CPU-only and their payload is the update descriptor, not a mesh. TripoSG (b) is the documented input-parity artifact above. Per-case values: `benchmarks/results/sppa_neural_flagship_wave.json`.

Method	Input	IoU	Triangles	Gen. time (ms)	VRAM (MB)	Payload (B)
SPPA-MVFit (ours)	top+side 96x96 telemetry masks	0.561	536	9.2	–	1,450
Generic-MVFit (context)	same masks	0.364 (-0.197)	1,280	11.4	–	1,433
Visual hull (context)	same masks	0.516 (-0.045)	51,432	0.2	–	32,768
TripoSR (a)	clean-crop shaded render	0.128 (-0.433)	28,145	384.1	1,869	1,468,998
TripoSR (b)	96x96 top mask	0.231 (-0.331)	46,053	379.4	1,870	2,454,199
Hunyuan3D-2mini-turbo (a)	clean-crop shaded render	0.157 (-0.403)	686,733	1350.9	4,591	10,513,018
Hunyuan3D-2mini-turbo (b)	96x96 top mask	0.171 (-0.390)	3,089,274	1915.5	4,747	46,494,183
TripoSG (a)	clean-crop shaded render	0.147 (-0.415)	697,881	9064.4	5,602	12,563,206
TripoSG (b)	96x96 top mask	0.002 (-0.559)	113,799	6281.8	5,409	2,049,198
Hunyuan3D-2 full (a)	clean-crop shaded render	0.148 (-0.413)	868,968	9264.3	5,650	13,037,869
Hunyuan3D-2 full (b)	96x96 top mask	0.177 (-0.385)	3,388,675	10005.6	5,936	50,848,377

S.3 Verifier-Gated Open-Label Probe

We evaluate the open-label route with 20 deliberately awkward labels that were not selected as the original paper demo classes. The set includes heavy equipment, vertical structures, containers, UAV-like objects, agricultural objects, and unsafe visual-artifact contexts. Table S4 reports the current verifier-gated recipe implementation. The metric and tag-only conditions use the same labels; metric inputs provide declared dimensions, while tag-only inputs use template priors.

Table S4: Uncached open-label SPPA probe. Family/proxy means non-fallback runtime output. Verified recipes are accepted only after deterministic checks over required roles, fallback-role absence, primitive budget, and dimensions.

Condition	Fallback	Family/proxy	Verified recipes	Critical mismatches
Verifier-gated recipes, metric	4/20	16/20	8/20	0
Verifier-gated recipes, tag-only	4/20	16/20	8/20	0

The remaining fallback cases in this probe are intentional: `dog statue`, `toy tractor`, `worker helmet`, and `cow silhouette` are visual-artifact or object-part contexts. This is a stronger claim than pure fallback but still narrower than arbitrary text-to-3D: SPPA accepts a generated proxy only when the recipe contract passes.

Quadruped profiles and open-label admission. For an animal-like object, SPPA does not infer anatomy from first principles: a label such as `cow`, `horse`, `dog`, or `quadruped` selects a morphology profile inside the same quadruped generator (torso proportions, head/neck placement, legs, tail), and the emitted proxy remains a low-polygon part graph, not a biological reconstruction. For a tractor or tractor-trailer the exact taxonomic class is not required: `vehicle`, `agricultural vehicle`, `truck`, or `trailer` normalize to a farm, heavy, or generic vehicle proxy. Weak or adversarial labels prefer the conservative fallback volume over an invented object.

S.4 Real-Image Probes, Replay, and Visual Evidence

Four real-image probes (cyclist, electrical tower, tractor, and tractor with trailer) exercise the dual-input contract of the main paper (Section 3.2). These user-supplied crops are real input evidence, not 3D ground truth. In the image path, a YOLOE-26s open-vocabulary detector reads the real image and produces approximate semantic evidence; in the tag-only path, SPPA receives a semantic tag/text input without using the image crop. In both cases the semantic text is routed through a SPPA normalizer before proxy selection, so the pipeline does not depend on YOLOE producing an exact object name. Table S5 summarizes the four probes, and Figure S2 (moved from the main text) contrasts the input crop, the detector evidence, the emitted SPPA proxy, and the neural baselines for each.

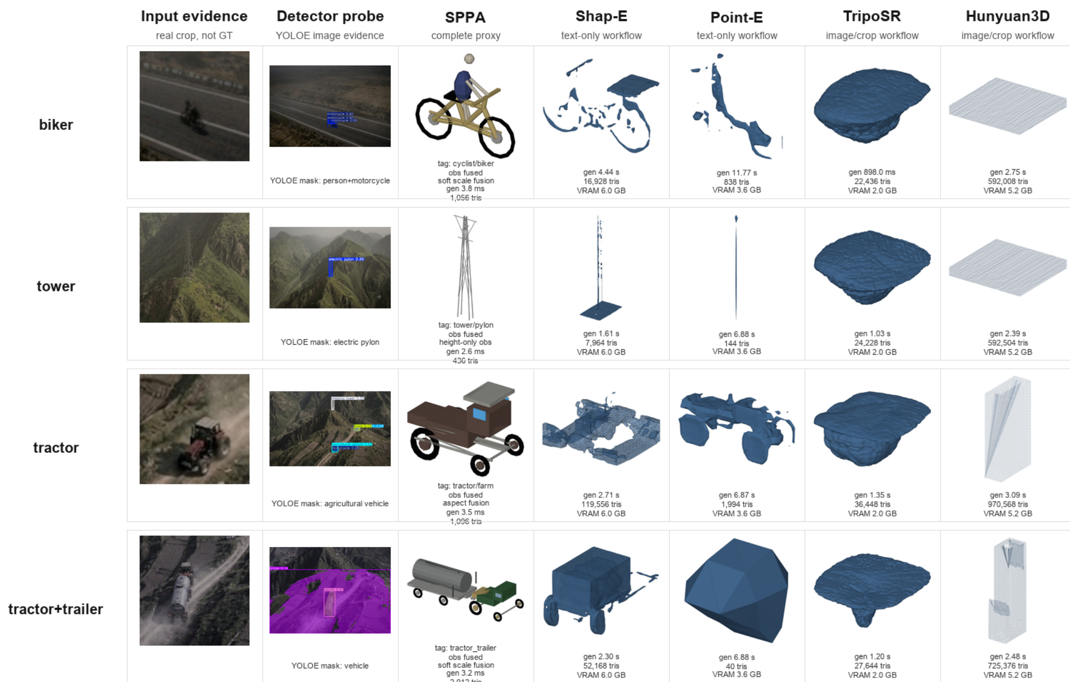

Figure S2: Real-image probes for biker, electrical tower, tractor, and tractor-with-trailer (rows), moved from the main text. Columns: input crop, YOLOE detector evidence, the emitted SPPA proxy, and four reproduced text/image-to-3D baselines (Shap-E, Point-E, TripoSR, Hunyuan3D) annotated with generation time, triangle count, and VRAM. Detector output is evidence, not ground truth; the neural columns receive a single image or crop, an input-modality mismatch shown for operating-point context, not a ranking.

Table S5: Real-image probe summary (audits in Tables S6–S9). YOLOE evidence is detector output, not ground truth. Raw dimensions come from mask/bbox projection under declared assumed-flight telemetry; SPPA dimensions are the constraint-fused proxy dimensions of the observation-fusion audit; wall time is the detector+observation route proxy build. No 3D ground truth exists for these images, so no IoU is reported here.

Case	YOLOE evidence (conf.)	SPPA family	Raw dims (m)	SPPA dims (m)	Tris	Wall (ms)
Cyclist	`person + motorcycle` (0.40)	`biker`	$3.59 \times 1.77 \times 1.85$	$1.85 \times 0.73 \times 1.85$	1,036	3.528
Tower	`electric pylon` (0.46)	`vertical_structure`	$12.77 \times 2.89 \times 28.00$	$5.60 \times 2.89 \times 28.00$	270	1.372
Tractor	`agricultural vehicle` (0.52)	`farm_vehicle`	$4.95 \times 4.32 \times 2.60$	$4.52 \times 2.38 \times 2.60$	880	2.477
Tractor-trailer	`vehicle` (0.47)	`articulated_vehicle`	$16.85 \times 6.21 \times 3.40$	$12.11 \times 3.76 \times 3.40$	1,904	2.963

Table S6: Real-image SPPA observation and replay audit per probe. Raw dimensions come from YOLOE mask/bbox projection under declared assumed-flight telemetry; SPPA dimensions are the constraint-fused proxy dimensions. Telemetry and height priors are declared replay parameters, so position and scale are scenario-relative, not measured ground truth.

Case	YOLOE evidence (conf.)	Replay proxy	Raw L/W/H (m)	SPPA L/W/H (m)	Gate	Local pose (m)	Tris
biker	`person + motorcycle` (0.40)	`biker`	3.59 x 1.77 x 1.85	1.85 x 0.73 x 1.85	soft low-conf fusion	$N = -3.67$, $E = -1.75$	1036
tower	`electric pylon` (0.46)	`vertical_structure`	12.77 x 2.89 x 28.00	5.60 x 2.89 x 28.00	height-only fusion	$N = -1.72$, $E = 0.56$	270
tractor	`agricultural vehicle` (0.52)	`farm_vehicle`	4.95 x 4.32 x 2.60	4.52 x 2.38 x 2.60	soft aspect fusion	$N = -8.45$, $E = -10.96$	880
tractor_trailer	`vehicle` (0.47)	`heavy_vehicle`	16.85 x 6.21 x 3.40	12.11 x 3.76 x 3.40	soft low-conf fusion	$N = -2.38$, $E = -7.61$	1904

Table S7: Real-probe semantic normalization and detector-plus-observation refinement (fusion of the normalization table and the refinement audit). The image-side detector evidence is not treated as exact ground truth; SPPA maps it to a proxy family, optionally refining weak detector labels when metric observation supports a safer conservative shape. The refinement route does not use the reviewed tag: it starts from YOLOE detector text and metric replay evidence; ms and Tris are its wall time and proxy size.

Probe	YOLOE image evidence	Detector-only family	Detector+observation family	SPPA reviewed text	Refinement rule	ms	Tris
Cyclist road image	`person + motorcycle`	`two_wheeled_rider` → `biker`	`two_wheeled_rider` → `biker`	`biker`	person+two-wheel	3.528	1036
Mountain tower image	`electric pylon`	`power_tower` → `vertical_structure`	`power_tower` → `vertical_structure`	`tower`	power infrastructure	1.372	270
Mountain tractor image	`agricultural vehicle`	`farm_vehicle` → `farm_vehicle`	`farm_vehicle` → `farm_vehicle`	`tractor`	farm vehicle	2.477	936
Tractor with trailer image	`vehicle`	`generic_vehicle` → `heavy_vehicle`	`articulated_vehicle` → `articulated_vehicle`	`tractor`+`trailer`	long-footprint metric refinement	2.963	1848

The dual-input benchmark of Table S8 freezes this two-path contract as an explicit artifact; it remains separate from any visual image-to-3D leaderboard because no 3D reference meshes exist for these real images.

Table S8: Frozen real-image and tag/text dual-input benchmark for the bounded SPPA systems claim. The image track uses YOLOE detector evidence and crops for image-to-3D baselines; SPPA additionally reports an observation-refined detector family when geometry supports a safer proxy. The tag/text track uses the reviewed word or phrase for SPPA and text-conditioned baselines. This is not a full visual image-to-3D leaderboard.

Case	YOLOE image input	SPPA family path	Image-track methods	Tag/text methods
biker	`person + motorcycle` (0.40)	detector: `two_wheeled_rider` → `biker`; obs: `two_wheeled_rider` → `biker`; text: `biker` → `biker`	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA
tower	`electric pylon` (0.46)	detector: `power_tower` → `vertical_structure`; obs: `power_tower` → `vertical_structure`; text: `tower` → `tower`	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA
tractor	`agricultural vehicle` (0.52)	detector: `farm_vehicle` → `farm_vehicle`; obs: `farm_vehicle` → `farm_vehicle`; text: `tractor` → `tractor`	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA
tractor+trailer	`vehicle` (0.47)	detector: `generic_vehicle` → `heavy_vehicle`; obs: `articulated_vehicle` → `articulated_vehicle`; text: `tractor_trailer` → `tractor_trailer`	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA

Figure S3 and Table S9 isolate the same SPPA generator under three input contracts: text-only (reviewed phrase), detector+metric (YOLOE text plus declared replay scale), and visual (agnostic image-space cues that may only condition existing roles). Visual evidence causes a measured geometry delta in all four cases at an additional cost of 20–40 triangles.

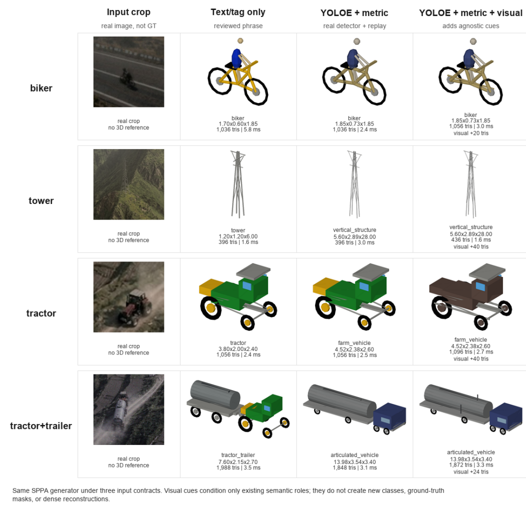

Figure S3: SPPA input-mode ablation on the four real-image probes: text-only, YOLOE+metric, and YOLOE+metric+visual contracts through the same deterministic generator. An ablation of SPPA evidence handling, not ground-truth 3D evaluation.

Table S9: SPPA input-mode comparison with bounded evidence-channel coverage. All rows call the same deterministic generator. Ch. counts the audited evidence channels consumed (semantic normalization, metric replay, visual role cues, yaw gating, observed material color); it is not a 3D quality score. All rows pass the lightweight runtime budget.

Case	Mode	Semantic	Dims L/W/H (m)	Visual tris	Δ vs metric	Ch.	Wall (ms)	Tris
biker	text	biker	1.70/0.60/1.85	-	yes	1	5.8	1036
biker	det+metric	biker	1.85/0.73/1.85	-	no	2	2.4	1036
biker	det+metric+visual	biker	1.85/0.73/1.85	20	yes	4	3.0	1056
tower	text	tower	1.20/1.20/6.00	-	yes	1	1.6	396
tower	det+metric	vertical	5.60/2.89/28.00	-	no	2	3.0	396
tower	det+metric+visual	vertical	5.60/2.89/28.00	40	yes	4	1.6	436
tractor	text	tractor	3.80/2.00/2.40	-	yes	1	2.4	1056
tractor	det+metric	farm	4.52/2.38/2.60	-	no	2	2.5	1056
tractor	det+metric+visual	farm	4.52/2.38/2.60	40	yes	5	2.7	1096
tractor+trailer	text	tractor+trailer	7.60/2.15/2.70	-	yes	1	3.5	1988
tractor+trailer	det+metric	articulated	13.98/3.54/3.40	-	no	2	3.1	1848
tractor+trailer	det+metric+visual	articulated	13.98/3.54/3.40	24	yes	4	3.3	1872

The refinement route of Table S7 does not use the reviewed tag: in the tractor-with-trailer case the system refines `vehicle` only to `articulated_vehicle`, never to the reviewed `tractor_trailer` class.

Metric replay and replay verifier. The metric proxy replay injects declared UAV pose, AGL, camera mount, and family height priors as scenario parameters; it is a real-image test of the observation-to-descriptor path, not a measured flight-localization result. A replay verifier checks the native-mask path: the selected YOLOE detections include native mask polygons with 12, 35, 33, and 54 points, projected into candidate oriented footprints before the constraint-fusion gate. This verifies the detector-evidence data path, not mask correctness against ground truth.

Visual bridge and connector regressions. An experimental agnostic image-space probe extracts generic cues (round-pair, line-structure, principal-axis, mask-envelope) from real crops and unlabeled masks; the descriptor consumes them only in a role-conditioned way, without creating new classes or topology. Connector regressions guard against disconnected-part proxies, and an anti-shortcut battery mutates labels, case identities, mask ordering, side channels, mirror transforms, and photometric variants; primary decisions pass in every row. These are plumbing checks, not evidence of universal primitive recovery; full tables and grids are retained in the artifact package (`benchmarks/results/`).

S.5 Role-Preservation Diagnostic

The implemented diagnostic asks whether SPPA preserves semantic part roles while adapting dimensions. For four synthetic vehicle variants, median dimension error was 0.3926 for fixed templates, 0.0000 for trivial global scaling, and 0.0223 for SPPA parametric vehicles; global scaling fits the outer box by deforming all parts together and is not a win condition. The supported claim is narrower: role-assigned parts can follow different update rules. A short/long truck invariance check makes this concrete: increasing vehicle length from 5.2 m to 8.2 m kept cab and tire scale deltas at 0.0, increased cargo length by 2.925 m, and increased tire count from 6 to 8. A calibrated-mask variant over 64 synthetic rectangle masks exercises the same path from observed footprint evidence (contract regression, artifact package).

S.6 Controlled Contract Regressions (Note)

Two controlled benchmarks verified the observation-to-descriptor plumbing as round-trip regressions under a known synthetic camera: a six-case contract benchmark (label-only prior versus metric-observation path) and a 162-mask bbox-versus-silhouette projection benchmark. Against self-generated synthetic references the metric paths recover the supplied dimensions exactly; these are scale-adaptation plumbing checks, not accuracy results on real detector masks, and the tables are retained in the artifact package (`benchmarks/results/`).

S.7 Superseded Substitutability Stress Test (Note)

An early substitutability stress comparison (July 2026; six ad-hoc local object cases; Shap-E, Point-E, TripoSR, and Hunyuan3D-2mini Turbo under a constrained local profile) is superseded for external-generator evidence by the measured neural wave of Section S.2 and is retained only in the artifact package. Its single durable observation is qualitative: generators that receive appropriate visual evidence produce richer meshes, at GPU cost and without a track-update contract.

S.8 Descriptor and Unreal Runtime Tests

Descriptor record contents (moved from the main text). A `SPPA-DESC-0.2` record stores resolver source, match type, fallback reason, scale source, mask footprint evidence, yaw ambiguity, shape confidence, material source, part list, primitive counts, and scheduler metadata; a `SPPA-UPD-0.2` record separates create/regenerate payloads from pose/no-op/shape updates, and `descriptor_id` acts as a topology-cache identity so that a pose update never looks like a new mesh descriptor.

In the descriptor/update replay benchmark, exact and keyword labels resolved to known finite archetypes and unsupported labels produced unknown fallback descriptors. Descriptor build with parts over 81,568 threshold-expanded rows had P50/P95/P99 of 391.8/515.0/671.2 μs (maximum observation 360.050 ms); scheduler decisions had P50/P95/P99 of 9.2/14.2/19.2 μs, and update-packet builds of 16.8/75.3/91.0 μs. Contract validation passed 453 descriptor and update-packet checks. These are Python contract measurements, not Unreal frame time.

A preliminary link-budget model separates byte accounting from radio-link validation: with no-op packets suppressed, measured SPPA JSON update traffic was 25.8–37.4 kB/s (78.5–94.0 kB/s emitting every packet), between a modeled binary box stream (about 3.27 kB/s) and modeled compressed video (250/625/1250 kB/s at 2/5/10 Mbps).

An adversarial resolver audit over 12 compound or misleading labels forced visual-artifact and equipment-only contexts (`toy truck`, `vehicle shadow`, `dog statue`, `person poster`, `worker helmet`) to the unknown fallback, while compound family labels resolved to reviewed archetypes. A complementary offline recipe-review gate accepted `dump truck` as a reviewed heavy-vehicle mapping, allowed `unlisted obstacle` only as fallback, and rejected `person poster`, `delivery drone`, and `construction crane` for false semantics, unreviewed roles, or runtime-LLM dependence.

Unreal tests exercised the native backend in three steps: an actor-level regression over a real `SPPA-DESC-0.2` fixture (12 parts, material and uncertainty tags, pose updates accepted, mismatched or invalid packets rejected), a component replay using the `/api/ui/data` payload shape, and a local-loopback HTTP replay, followed by Editor-Cmd and packaged recorded-payload runs through the same component entry point. This remains an internal short-stream replay, not live telemetry, VR, or operator validation.

Table S10: Selected Unreal evidence that constrains the SPPA claim. Values are preliminary and do not include VR headset, live-network, broad curated-asset validation, or phase-aligned GPU-profiler validation. Auxiliary regression checks are kept in the artifact package, not in this summary table.

Test	Result	Main boundary	Claim impact
Backend switch smoke	Same obstacle input selects assets, actor proxies, or HISM proxies; default remains assets	Editor smoke, not operator workflow	SPPA is optional, not a forced replacement
Editor-Cmd recorded replay	27/27 payload applications accepted across asset, actor-proxy, and HISM backends; SPPA emit-all P95 apply: 1.614/3.805/4.049 ms	component ingestion under Editor-Cmd/Null-RHI	recorded streams reach all backend routes
Packaged recorded replay `20260703T024800Z`	27/27 payload applications accepted; frame P95 assets/proxy/HISM: 16.551/6.531/4.809 ms; apply P95: 2.566/5.670/3.246 ms	one short replay, no live network, no phase-aligned GPU profiler, no headset	recorded streams run in packaged executable
Packaged synthetic 100-object replay	actor-proxy pose/shape frame P95: 17.814/19.256 ms	synthetic desktop obstacles; misses 60 FPS P95	desktop feasibility only, not VR readiness
Packaged HISM scaling `20260703T033655Z/033840Z`	11 active draw groups; 100-object no-CSV rerun shape frame P95: 37.126 ms; 250/500 shape P95: 94.018/186.113 ms	synthetic desktop obstacles; shape updates miss 30 FPS P95 at 100+ objects	batching helps draw pressure, but dense shape updates remain limiting
Packaged HISM partial updates `20260703T040602Z`	10% pose and 10% shape changed-track schedule; 500-object pose/shape frame P95: 16.528/29.787 ms with 11 draw groups	synthetic scheduler rate, not flight-derived; dense all-object updates still fail	partial-update contract supports changed-only scheduling, not dense all-track updates
Packaged cow/tower asset subset	project assets vs. SPPA shape frame P95: 6.789 vs. 22.445 ms	narrow two-label subset	curated assets remain better when available

The packaged evidence consists of synthetic density replay, HISM batching diagnostics, and a short recorded-payload replay. The baseline packaged replay, `20260702T122743Z`, used three repetitions, 30 warmup frames, and 120 measured frames per phase. At 100 synthetic obstacles, `semantic_proxy` used 432 components, about 208,864 estimated triangles, and 432 estimated draw calls; pose and shape frame P95 were 17.814 ms and 19.256 ms, passing a 33.3 ms P95 desktop frame budget but missing a 16.7 ms P95 budget; phase-aligned GPU-profiler counters were not captured. The recorded-payload run `20260703T024800Z` replayed three externally generated JSONL streams (common asset-compatible, SPPA emit-all, and SPPA changed-only) through the same backend switch; all 27 applications were accepted, with the HISM route at median 6 active draw groups and 4.809 ms frame P95 against 31 components and 6.531 ms for the actor proxy.

Across the HISM sequence, batching reduced the 100-object component and estimated draw-call count from 432 to 11 active draw groups. Dense shape updates are the remaining limit: even after batching, 100-object shape-stream frame P95 stayed above the 30 FPS budget (37.126 ms in the isolated no-CSV rerun) and rose to 94.018/186.113 ms at 250/500 objects. The HISM path was therefore extended with an explicit `partial_obstacle_update` contract—unmentioned entities are retained, mentioned entities replace their old instance keys before new parts are upserted, and the backend verifier checks 6/6 expected live/HISM instances when only one of two entities is updated. Under a changed-track schedule with pose and shape updates each limited to 10% of objects, 500-object pose/shape frame P95 was 16.528/29.787 ms with 11 draw groups and no measured hitches above 33 ms. Stable dense shape-update performance is not demonstrated even at 100 objects, and the 10% rate is a synthetic scheduler setting rather than a flight-derived one.

A second packaged run adds a narrower real-asset baseline for the two labels with reviewed project static meshes in the repository, `cow` and `tower`. At 50 objects, `project_assets` used 50 static-mesh components and about 167,300 estimated triangles; `semantic_proxy` used 200 components and about 113,200 estimated triangles, with shape-stream P95 6.789 ms versus 22.445 ms for SPPA. Curated assets are preferable when an appropriate asset exists and loads cheaply; SPPA's intended gap is uncovered or mismatched semantic telemetry, not replacement of available high-quality assets.

Figure S4 (moved from the main text) plots the dense packaged-runtime scaling series behind the corresponding row of the condensed runtime table in the main text.

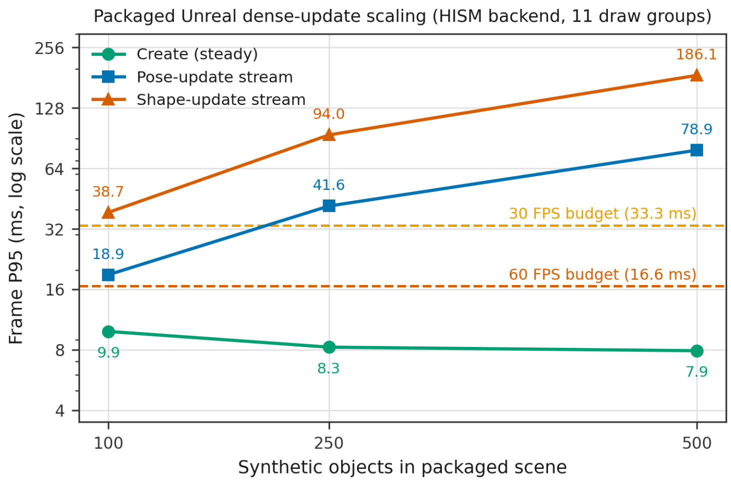

Figure S4: Packaged-runtime scaling (HISM backend, dense-update streams), moved from the main text: frame P95 versus object count for the create, pose-update, and shape-update streams. The 33.3 ms line marks a desktop 30 FPS budget; dense shape updates exceed it already at 100 objects and dense pose updates between 100 and 250, while the sparse 10% changed-track schedule of the condensed runtime table in the main text stays below it at 500 objects. Run note: the plotted series comes from the original scaling run `20260703T033655Z` (CSV logging active; 100-object shape frame P95 38.7 ms), whereas the condensed runtime table in the main text and the table above quote the isolated no-CSV rerun `20260703T033840Z` (37.126 ms). The two runs differ only in absolute milliseconds; dense shape updates exceed the 33.3 ms budget from 100 objects in both, so the conclusion is unchanged.

S.9 Sealed and Post-Hoc Analysis Tables

Full tables behind the figures of Sections 4.1–4.5 of the main paper, plus material moved from the main text during the mission reframe (the secondary paired differences, the surface metrics, the OBB baseline detail, the fitting-sequence and external-gallery figures, and the real-stream localization analysis). Table S11 belongs to the confirmatory run, as does the prespecified observation-condition table, which now appears in the main text (Table 6); the rest are post-hoc exploratory analyses on the released artifacts and change nothing about H1 or its interval.

Table S11: Paired clean-mask mean IoU differences of SPPA-MVFit minus Generic-MVFit by morphology family and source stratum. All twelve cells are positive; the smallest gain is the compact-vehicle implicit-OOD cell. The equal-stratum aggregate is the primary estimate of Table 2.

Family	CSG-ID Δ	Implicit-OOD Δ	Mean Δ
articulated_vehicle	0.113	0.125	0.119
branching_vertical	0.200	0.210	0.205
compact_vehicle	0.101	0.043	0.072
lattice_tower	0.175	0.163	0.169
quadruped	0.205	0.189	0.197
rider_cycle	0.458	0.299	0.379
Equal-stratum aggregate	0.209	0.172	0.190

Table S12: Paired clean-mask mean IoU differences of SPPA-MVFit minus each comparator (moved from the main text; stratified actor bootstrap 95% intervals, per-comparison and unadjusted across secondary endpoints). Visual hull is a high-complexity geometry reference, not a lightweight runtime actor.

Comparator	Mean Δ IoU	95% CI low	95% CI high
Generic-MVFit (H1)	0.190	0.181	0.199
SPPA text-only	0.130	0.116	0.144
Visual hull	0.036	0.027	0.044
Ellipsoid	0.216	0.207	0.224
Capsule	0.233	0.224	0.241
Axis-aligned box	0.310	0.301	0.318
Billboard	0.384	0.375	0.393

Figure S5 (moved from the main text) shows one fitted example of the confirmatory fitting path, and Figure S6 (moved from the main text) the two representative outcomes of the external sanity check of Section 4.7.

Oriented bounding box baseline (moved from the main text). The primitive baselines in the main benchmark are axis-aligned. A rotation-aware oriented bounding box (top-mask minimum-area rectangle with gap-midpoint refinement, z extent from the side mask) scores 0.252, practically identical to the axis-aligned box (0.248; paired

Table S13: Surface metrics on the clean condition (n=240), moved from the main text. F-score at a 1.5-voxel threshold against ground-truth surfaces; paired differences versus SPPA-MVFit with stratified bootstrap 95% intervals.

Method	F-score@1.5 vox	Δ vs SPPA	95% CI
SPPA-MVFit	0.831	—	—
Generic-MVFit	0.560	−0.271	[−0.280, −0.262]
Visual hull	0.799	−0.032	[−0.040, −0.023]
SPPA text-only	0.706	−0.125	[−0.140, −0.110]

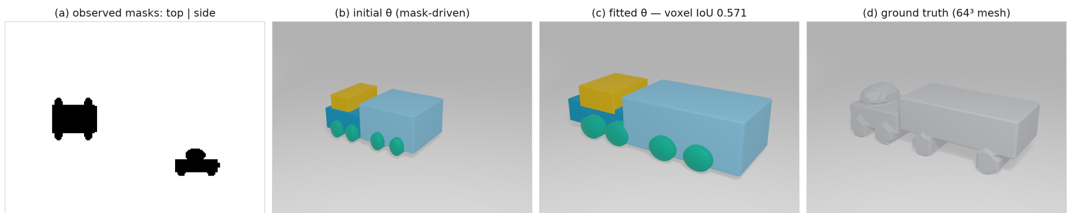

Figure S5: Fitting of SPPA-MVFit on a held-out actor (moved from the main text): (a) the observed silhouette pair, (b) initialization from the observed mask extents, (c) the converged eight-part proxy, and (d) the ground-truth voxel mesh.

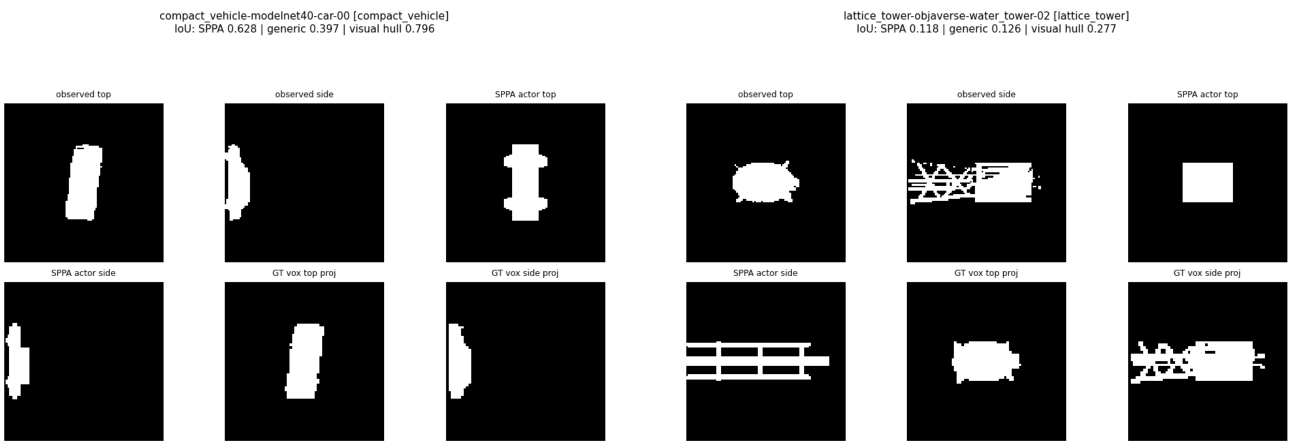

Figure S6: External sanity check, qualitative (moved from the main text). Left: a ModelNet40 car where the family graph helps (SPPA 0.628 vs generic 0.397). Right: an Objaverse water tower whose horizontal tank mismatches the vertical tower template (SPPA 0.118 vs generic 0.126)—the template–instance mismatch failure mode. Panels show observed masks, SPPA actor projections, and ground-truth voxel projections.

Table S14: Graph-by-fitting 2 × 2 decomposition (clean condition, n=240, mean voxel IoU). The Generic no-fit cell was computed after the confirmatory run through the same builder and evaluation, validated bit-exactly against the confirmatory metrics; the remaining cells match confirmatory values. Fitting-effect and per-stratum estimates are in Table S20.

Graph prior	No fit (θ_0)	MVFit	Fitting effect
Generic graph	0.180	0.367	0.187
SPPA family graph	0.427	0.557	0.130
Graph effect (SPPA − Generic)	0.248	0.190	−0.058

Table S15: Drop-one-family re-analysis of the primary paired difference (SPPA-MVFit minus Generic-MVFit, clean condition). Every leave-one-out aggregate stays far above the prespecified +0.030 margin.

Excluded family	Δ IoU	95% CI	n
None (all families)	0.190	[0.181, 0.199]	240
compact_vehicle	0.214	[0.203, 0.224]	200
articulated_vehicle	0.204	[0.194, 0.215]	200
quadruped	0.189	[0.178, 0.199]	200
branching_vertical	0.187	[0.178, 0.196]	200
lattice_tower	0.194	[0.184, 0.204]	200
rider_cycle	0.152	[0.143, 0.162]	200

Table S16: Wrong-family analysis (clean condition, n=240, mean voxel IoU). Rows: true family; columns: family token supplied to the fitter. Bold diagonal: correct token (= confirmatory SPPA-MVFit). Last row: mean over the five wrong tokens per column.

True family ↓ / token →	compact_vehicle	articulated_vehicle	quadruped	branching_vertical	lattice_tower	rider_cycle
compact_vehicle	**0.716**	0.560	0.462	0.063	0.165	0.213
articulated_vehicle	0.581	**0.590**	0.386	0.177	0.124	0.167
quadruped	0.349	0.269	**0.662**	0.281	0.155	0.139
branching_vertical	0.154	0.182	0.299	**0.533**	0.195	0.134
lattice_tower	0.127	0.084	0.117	0.164	**0.313**	0.134
rider_cycle	0.116	0.091	0.115	0.060	0.084	**0.530**
Wrong-token row mean	0.292	0.287	0.239	0.193	0.125	0.093

Table S17: View ablation of SPPA-MVFit (clean condition, n=240, correct family token). Single-view arms use an isotropic prior for the unobserved scale, documented before measurement.

Fit variant	Mean IoU	Δ vs dual	CI95 low	CI95 high
Dual-view (sealed SPPA-MVFit)	0.557	—	—	—
Top-only fit	0.458	-0.100	-0.112	-0.088
Side-only fit	0.545	-0.012	-0.021	-0.003
Stratum csg_id: dual 0.551, top-only 0.424, side-only 0.551				
Stratum implicit_ood: dual 0.564, top-only 0.491, side-only 0.540				

Table S18: Generic-graph sensitivity (clean condition, n=240, mean voxel IoU). Variants registered in writing before measurement; G1 re-runs the original generic and is validated bit-exactly against the confirmatory metrics. Gap closed is relative to the SPPA=100% endpoint.

Generic graph variant	Mean IoU	Δ SPPA − variant	CI95	Gap closed
G1 sealed generic (control)	0.367	0.190	—	—
G2 box/cylinder chassis	0.339	0.219	[0.210, 0.228]	-15.2%
G3 vertical stack + legs	0.282	0.275	[0.266, 0.285]	-45.0%
G4 slot-wise family mean	0.330	0.228	[0.219, 0.236]	-19.8%
SPPA-MVFit (sealed)	0.557	0.000	—	100.0%

Table S19: All-method means under the prespecified observation conditions (evaluated with the confirmatory run; n=240 per condition).

Method	Clean	Mild morph.	Moderate morph.	Partial occl.	Mask corrupt.
SPPA-MVFit	0.557	0.512	0.418	0.545	0.555
Generic-MVFit	0.367	0.349	0.299	0.355	0.367
SPPA text-only	0.427	0.427	0.427	0.427	0.427
Axis-aligned box	0.248	0.264	0.264	0.258	0.248
Ellipsoid	0.342	0.343	0.324	0.353	0.342
Capsule	0.325	0.334	0.316	0.336	0.325
Billboard	0.173	0.153	0.122	0.148	0.170
Visual hull	0.522	0.470	0.356	0.397	0.510

Table S20: Post-hoc exploratory effect estimates for the graph-by-fitting 2×2 decomposition (clean condition; pooled and per stratum).

Effect	Estimate	95% CI	n
Graph effect at no-fit	0.248	[0.234, 0.261]	240
Fitting effect, generic graph	0.187	[0.182, 0.193]	240
Fitting effect, SPPA family graph	0.130	[0.117, 0.143]	240
Total (SPPA-fit − Generic-fit)	0.190	[0.181, 0.199]	240
Interaction (fitting×graph)	−0.058	–	240
By stratum			
Graph effect at no-fit (CSG-ID)	0.239	[0.220, 0.258]	120
Fitting effect, generic graph (CSG-ID)	0.177	[0.168, 0.186]	120
Fitting effect, SPPA family graph (CSG-ID)	0.147	[0.124, 0.169]	120
Total (SPPA-fit − Generic-fit) (CSG-ID)	0.209	[0.192, 0.225]	120
Graph effect at no-fit (Implicit-OOD)	0.256	[0.238, 0.273]	120
Fitting effect, generic graph (Implicit-OOD)	0.197	[0.191, 0.204]	120
Fitting effect, SPPA family graph (Implicit-OOD)	0.113	[0.098, 0.128]	120
Total (SPPA-fit − Generic-fit) (Implicit-OOD)	0.172	[0.162, 0.181]	120

Table S21: Post-hoc exploratory Chamfer distance by observation condition (all methods, n=240); paired differences versus SPPA-MVFit with 95% bootstrap intervals.

Method	Clean	Mild morph.	Moderate morph.	Partial occl.	Mask corrupt.	Δ vs SPPA	95% CI
SPPA-MVFit	0.008	0.009	0.012	0.008	0.008	—	—
Generic-MVFit	0.016	0.017	0.021	0.017	0.016	0.008	[0.008, 0.008]
SPPA text-only	0.012	0.012	0.012	0.012	0.012	0.004	[0.004, 0.005]
Axis-aligned box	0.026	0.026	0.120	0.025	0.026	0.018	[0.018, 0.019]
Ellipsoid	0.018	0.018	0.115	0.018	0.018	0.010	[0.010, 0.011]
Capsule	0.020	0.020	0.116	0.019	0.020	0.012	[0.011, 0.012]
Billboard	0.017	0.018	0.114	0.018	0.019	0.009	[0.009, 0.009]
Visual hull	0.009	0.011	0.111	0.011	0.011	0.001	[0.001, 0.001]

Table S22: Post-hoc exploratory worst SPPA-MVFit cases (clean condition, voxel IoU at 64^3). The five sub-threshold cases (IoU < 0.25) are all `lattice_tower` CSG-ID; see the failure analysis in Section 4.2 of the main paper.

Case	Family	Stratum	IoU	Chamfer	Vol. err
test-csg_id-lattice_tower-017	lattice_tower	csg_id	0.147	0.010	1.364
test-csg_id-lattice_tower-007	lattice_tower	csg_id	0.148	0.010	1.400
test-csg_id-lattice_tower-011	lattice_tower	csg_id	0.173	0.009	1.076
test-csg_id-lattice_tower-018	lattice_tower	csg_id	0.182	0.012	1.244
test-csg_id-lattice_tower-008	lattice_tower	csg_id	0.239	0.008	0.228
test-implicit_ood-lattice_tower-015	lattice_tower	implicit_ood	0.251	0.006	0.379
test-csg_id-lattice_tower-000	lattice_tower	csg_id	0.257	0.007	0.449
test-implicit_ood-lattice_tower-013	lattice_tower	implicit_ood	0.257	0.007	0.610
test-csg_id-lattice_tower-005	lattice_tower	csg_id	0.261	0.009	0.005
test-implicit_ood-lattice_tower-009	lattice_tower	implicit_ood	0.265	0.008	0.856

Table S23: Post-hoc exploratory optimizer budget sweep (clean condition, n=240, correct family token). The adopted budget is 31 candidates; doubling it gains a non-significant +0.002 IoU for twice the wall time.

Candidate budget	Mean IoU	Mean ms	Median ms	p95 ms
11	0.528	3.574	3.537	4.992
21	0.547	7.590	7.540	9.355
31 (sealed)	0.557	12.566	12.393	14.179
61	0.560	24.034	23.642	27.690

Table S24: Post-hoc exploratory wrong-family paired comparisons (clean condition, n=240; stratified bootstrap 95% intervals).

Comparison	Mean diff	CI95 low	CI95 high
Wrong-token mean − correct token	-0.353	-0.362	-0.343
Wrong-token mean − Generic-MVFit	-0.162	-0.167	-0.158
Correct token − Generic-MVFit	0.190	0.181	0.199

difference -0.004, CI $[-0.005, -0.003]$): the benchmark actors are presented axis-aligned, so orientation freedom adds nothing, and SPPA-MVFit minus OBB is $+0.306$ (CI $[0.297, 0.314]$; Table S25). The box family of baselines, oriented or not, does not close the representation gap. The wider representation design space is mapped in Section S.1.

Table S25: Post-hoc exploratory oriented bounding-box baseline on the primary benchmark (n=240, sealed methods marked). The OBB fitted from the given world-metric box stays within 0.004 IoU of the axis-aligned box, so box orientation does not explain the family-graph effect.

Method	Mean IoU	Δ vs OBB	CI95
Axis-aligned box (sealed)	0.248	-0.004	[-0.005, -0.003]
OBB (this work)	0.252	—	—
Visual hull (sealed)	0.522	0.270	[0.264, 0.275]
SPPA-MVFit (sealed)	0.557	0.306	[0.297, 0.314]

Table S26: Post-hoc exploratory role-aware IoU on the CSG-ID stratum (n=120 actors, 920 matched slot–component pairs; slot-to-component mapping frozen in writing before computation). SPPA-MVFit reaches 0.319 role IoU versus 0.017 (cyclic) and 0.053 (random) shuffle controls.

Family	n pairs	Role coverage	Role IoU	Shuffle IoU (cyclic)	Shuffle IoU (random)
compact_vehicle	140	0.724	0.413	0.002	0.062
articulated_vehicle	160	0.614	0.285	0.013	0.046
quadruped	160	0.707	0.423	0.001	0.062
branching_vertical	140	0.304	0.187	0.078	0.066
lattice_tower	160	0.344	0.231	0.004	0.032
rider_cycle	160	0.569	0.372	0.006	0.052
Overall (csg_id)	920	0.545	0.319	0.017	0.053

Table S27: Post-hoc exploratory adversarial-family stratum (n=120; twenty actors per family, twelve documented violation types, design frozen before measurement). Clean columns re-run the same actors unviolated. Violating the structural prior cuts the paired SPPA advantage from +0.209 to +0.141 and produces statistical ties (leaning lattice, cascade crown); SPPA loses at actor level in 8.3% of cases.

Family	n	SPPA clean	SPPA adv	Gen clean	Gen adv	Δ_{adv}	CI95	SPPA<Gen
compact_vehicle	20	0.686	0.481	0.585	0.426	0.055	[0.027, 0.079]	10%
articulated_vehicle	20	0.513	0.465	0.400	0.370	0.095	[0.079, 0.111]	0%
quadruped	20	0.652	0.537	0.447	0.386	0.151	[0.124, 0.178]	0%
branching_vertical	20	0.547	0.343	0.348	0.246	0.098	[0.052, 0.145]	20%
lattice_tower	20	0.316	0.211	0.141	0.088	0.123	[0.067, 0.185]	20%
rider_cycle	20	0.591	0.464	0.134	0.141	0.323	[0.282, 0.362]	0%
Overall	120	0.551	0.417	0.342	0.276	0.141	[0.125, 0.157]	8%

Table S28: Post-hoc exploratory part-query task on the CSG-ID stratum (887 scorable role queries over 120 actors; query set, hull heuristics, and mapping frozen before measurement). F1 and normalized part-centroid error d_c by family. The hull heuristics cover but cannot select; SPPA's structural count is exact in 70% of actors versus 0% for the hull.

Family	n	SPPA F1	Gen F1	HullA F1	HullB F1	SPPA d_c	Gen d_c	HullA d_c	HullB d_c
compact_vehicle	140	0.555	0.246	0.139	0.123	0.038	0.088	0.272	0.277
articulated_vehicle	160	0.412	0.171	0.140	0.150	0.058	0.134	0.286	0.231
quadruped	143	0.598	0.238	0.123	0.064	0.037	0.082	0.265	0.280
branching_vertical	138	0.253	0.032	0.132	0.044	0.125	0.275	0.151	0.268
lattice_tower	156	0.308	0.037	0.086	0.092	0.047	0.367	0.150	0.231
rider_cycle	150	0.484	0.153	0.049	0.043	0.032	0.106	0.203	0.215
Overall	887	0.434	0.145	0.111	0.087	0.055	0.177	0.221	0.249

Real-stream localization analysis (moved from the main text). Localization on this stream is observation-bound, not fitter-bound: all methods sit at ≈33 m median 3D error, exactly on the identity line of footprint-center error versus anchor error, because the low-altitude stream (AGL 10–23 m) compresses monocular range; footprint IoU against a 5×5 m base is degenerate for every method. Table S29 reports the full per-method table including the localization and footprint columns, and Figure S7 the plan-view and error-identity panels behind them. Localization is outside the main paper's scope (position assumed given); it is retained here for completeness.

E10: measured mode routing and token validation (exploratory post-hoc). A candidate deployment prescription—route each detection between the footprint-only SPPA mode and a box-proxy mode using detector-token confidence or prior mismatch—was frozen as a protocol (`benchmarks/real_stream_wave/e10_protocol.md`) on 2026-07-19 *before* any routing outcome was computed, then measured on the 1,902 real-stream cases with the 2D reprojection IoU of the routed method as outcome (case-level paired bootstrap, 10,000 resamples, seed 20260719). Table S31 and Figure S8 report both halves. Routing is refuted on this outcome: every threshold policy collapses to always-proxy, the paired best-minus-always-proxy median difference is 0.0 $[0, 0]$, and the unreachable per-case oracle

Table S29: Real detector stream case study, full table including the localization columns (exploratory post-hoc, not sealed): 1,902 detections from a 1,394-frame flight over the georeferenced twin with a real custom detector, real telemetry, and exact tower-anchor ground truth. Localization and footprint columns: GT-matched subset (n=217; wrong detector tokens kept as a natural condition); reprojection and latency: all cases. All methods are observation-bound at ≈33 m; SPPA-MVFit trades 2D image overlap for 3D part structure (the stream has no 3D ground truth).

Method	Loc. err. 3D med. (m)†	Footprint IoU†	2D reproj. IoU med. [P25, P75]	Latency (ms)
SPPA-MVFit (this work)	32.8	0.000	0.298 [0.218, 0.392]	11.8
Generic-MVFit	32.8	0.000	0.423 [0.394, 0.448]	13.4
OBB	32.7	0.000	0.446 [0.398, 0.492]	0.22
AABB	32.7	0.000	0.330 [0.299, 0.357]	0.25
Visual hull (non-semantic)	32.7	0.000	0.446 [0.398, 0.493]	0.50
Capsule	32.7	0.000	0.445 [0.422, 0.466]	1.31

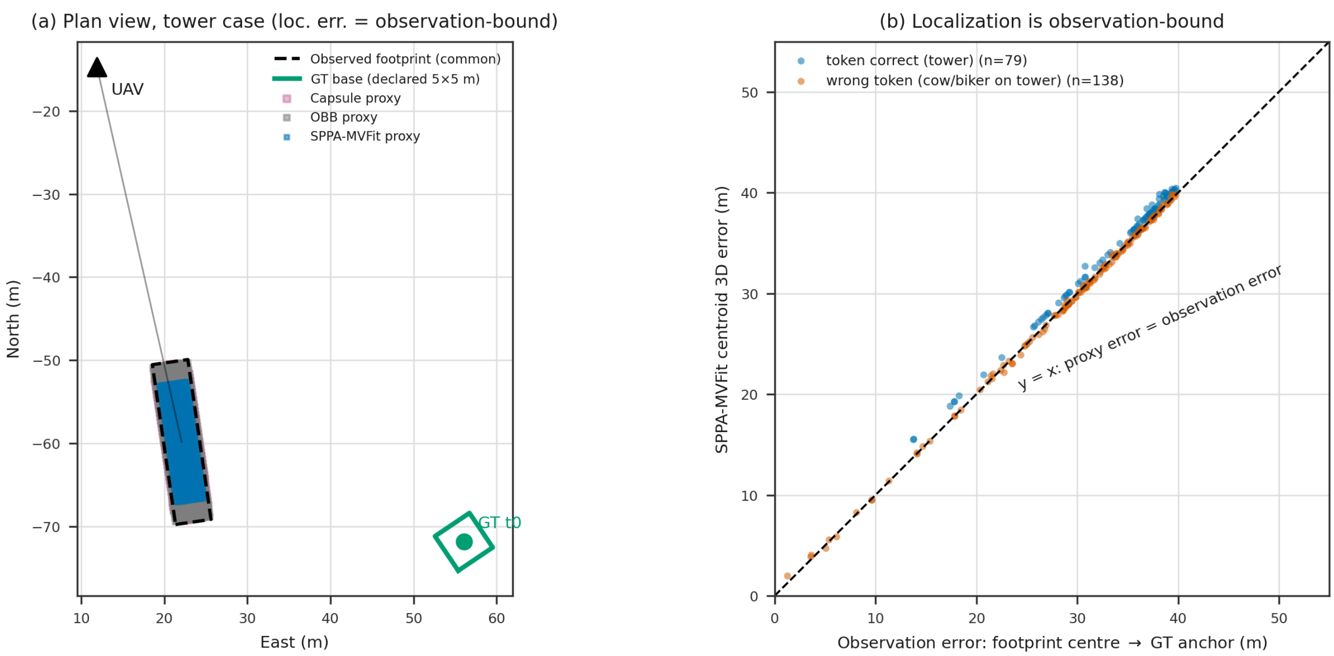

Figure S7: Real-stream localization panels (moved from the main text). (a) Plan view of a tower case: the observed footprint (dashed) and all fitted proxies sit ≈35 m from the exact anchor—observation-bound. (b) Fitted-centroid error versus observation error: the identity line holds for all methods.

Table S30: Real-stream token arm (SPPA-MVFit only; exploratory post-hoc): the 138 GT-matched cases where the real detector emitted a wrong family token (114 cow→tower, 24 biker→tower). Refitting with the correct token collapses the 2D reprojection fit—the correct prior cannot explain the misleading evidence that triggered the wrong label.

Token condition	n	2D reproj. IoU med. [P25, P75]
Real detector token (wrong)	138	0.381 [0.303, 0.410]
Correct-token refit (counterfactual)	138	0.025 [0.023, 0.034]

gains only +0.006 over always-proxy while picking SPPA in 17.1% of cases. The token-validation arm returns a usable signal instead: on the 217 GT-matched cases (138 wrong token, 79 correct), −confidence detects wrong tokens with AUC 0.847, while the declared-direction mismatch score points the other way (AUC 0.000)—wrong tokens show *lower* prior mismatch, so −mismatch separates perfectly (AUC 1.000, as does the secondary raw-height score). The mechanism is monocular: hallucinated cows/bikers receive small observed heights (≈1–2 m) that fit their wrong family priors suspiciously well, whereas true towers suffer severe monocular height underestimation (≈5 m observed against the 25 m prior, mismatch ≈1.6). The derived family heights behind the signal were frozen in the protocol (lattice_tower 25.0 m, quadruped 1.5 m, rider_cycle 1.8 m). The perfect separation is measured on 217 matched cases of a single stream and is not claimed beyond it.

Table S31: E10 measured mode routing (exploratory post-hoc; protocol frozen pre-outcome). 1,902 real-stream cases; outcome: 2D reprojection IoU of the routed method; case-level paired bootstrap, 10,000 resamples, seed 20260719. The oracle is an unreachable upper bound. Full policy grid: `benchmarks/real_stream_wave/e10_routing.json`.

Policy	Rule	Cov. → SPPA	2D reproj. IoU med. [95% CI]
Always SPPA-MVFit	all -> SPPA	100.0%	0.298 [0.290, 0.305]
Always proxy (OBB)	all -> proxy	0.0%	0.446 [0.442, 0.450]
Oracle (upper bound)	per-case argmax (upper bound)	17.1%	0.452 [0.448, 0.456]
Best conf. threshold	conf $<$ 0.7700 -> proxy	0.0%	0.446 [0.442, 0.450]
Best mismatch threshold	mismatch $>$ 0.0008 -> proxy	0.1%	0.446 [0.442, 0.450]
Best combined (AND)	conf $<$ 0.7700 AND mismatch $>$ 0.0008 -> proxy	0.1%	0.446 [0.442, 0.450]
Best combined (OR)	conf $<$ 0.7700 OR mismatch $>$ 0.0008 -> proxy	0.0%	0.446 [0.442, 0.450]

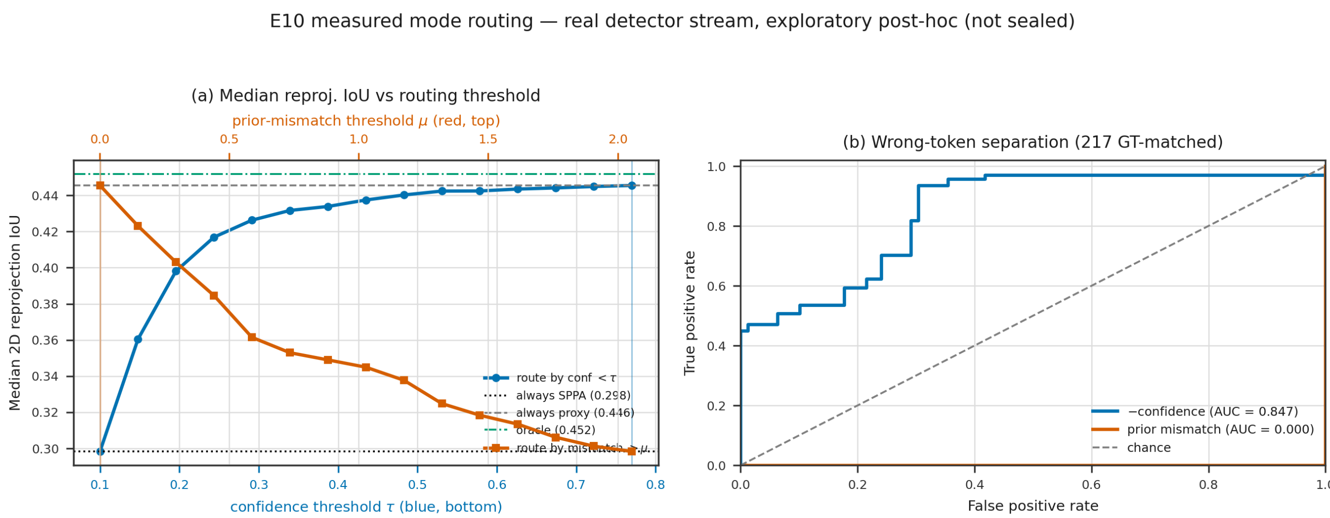

Figure S8: E10 measured mode routing and token validation. (a) Median 2D reprojection IoU against the routing threshold for both candidate signals, with always-SPPA, always-proxy, and oracle reference lines: both curves collapse to the always-proxy endpoint. (b) Wrong-token separation on the 217 GT-matched cases: −confidence reaches AUC 0.847; the declared-direction prior-mismatch curve collapses to AUC 0.000—the signal points opposite to intuition, so −mismatch separates perfectly (AUC 1.000).

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